

- Q. Why do we search along a segment? - A. Because Fibonacci and Golden section methods are one dimensional search techniques, so we need a 1D search interval.

Optimization Techniques

Due Date March 12, 2026

Assignment (1) 12 points

unimodal == 'has one min/max'.

According to chatgpt, this is the Golden Section method.

- (1) It is required to find the minimum of the unimodal function $F = 15e^{-x} + 2e^y$ along the segment joining the two points (1, 1) and (5, 4) within an interval of length 0.1. Use both Fibonacci search and Golden section methods to locate the minimum.
- (2) It is required to find the maximum of the unimodal function $F = \ln x - 2x^2$ within the interval (0, 1). Use the method which requires the minimum number of function evaluations to locate the maximum in an interval of length 0.05.
- (3) Use Fibonacci search to find the point of minimum energy $E = 14x^2 + y^2 + z^2 - 4xy - 6xz - 3x$ along the segment joining the two points (1, 2, 4) and (3, 5, 8) with an error $\varepsilon \leq 0.05$. Comment on the relation between the gradient at the point of minimum and the segment direction.
- (4) Consider the problem: Minimize $F = (x + 2y - 4)^2 + (3x - 4y - 2)^2$. Starting at (0,0), find the minimum along the steepest decent direction at (0, 0) using Powell's quadratic interpolation and $\delta = 1$. How many quadratic interpolations are expected to lead to the minimum in this direction?
- (5) Construct three simplices of the simplex method to find the minimum of $F = 5x^2 - 4xy + 10y^2 - 7x$ using the initial simplex with vertices $\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0.5 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0.5 \end{bmatrix}$. Show the steps on a graph paper.
- (6) Carry out three steps to find three new base points of pattern search to Minimize $F = x_1^2 x_2^2 + (x_1 - 2)^2 (x_2 - 3)^2$ starting at $\mathbf{x} = (1, 1)$, $\delta = 0.2$ for both directions. Show the steps taken on graph paper.
- (7) If the function $F(\mathbf{x})$ is first order differentiable then the statements " $F(\mathbf{x})$ is convex" and the inequality " $F(\mathbf{x}) \geq F(\mathbf{x}_0) + (\mathbf{x} - \mathbf{x}_0)^T \nabla F(\mathbf{x}_0)$, for all $\mathbf{x}$ " are equivalent. Moreover, if $F(\mathbf{x})$ is also second order differentiable then convexity is equivalent to the statement that "The Hessian matrix of $F(\mathbf{x})$ is positive definite for all $\mathbf{x}$ ". Notice that $\mathbf{x}$ is a vector and that equivalence means if and only if.

There's no interval given, just a starting point $\mathbf{x}_0$. Therefore, we should use "Davies, Swann and Campey"

It is required to prove these equivalences related to convexity.

MATLAB ASSIGNMENT

Write a code to apply the simplex method for case of n variables (Nelder and Mead).

Input: Number of variables n , the starting point $\mathbf{X}$ and a step δ used to construct the initial simplex by equal steps along the variable directions. Two stopping accuracies should be given as well for both function deviation and size of final simplex.

- i- Make your program in partitions: Initialization, Reflection, Expansion, Contraction and Stopping.
- ii- Test your program with problem (5) above. Stopping accuracy in F and for simplex size is 10^{-2} .
- iii- Minimize

$$F = (x_1 - 3)^2 + (x_2 - x_1^2)^2 + (x_2 - x_3)^2$$

starting at (2, 8, 6) and step $\delta = 0.5$. Stopping accuracy in F and for simplex size is 10^{-2} .